

Curation Reviewer Guide

1947 King USDA Repellent Dataset

Plain-language instructions for reviewing chemical names and structures. No programming or chemistry software experience required.

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1. Quick start - what you do

Open the curation web app (link from your team lead). Enter your name in the sidebar. Work through Queue 1 (NEEDS_REVIEW) first - these are the highest-priority rows. For each row you will see the 1947 archive name, other names, flags, and a structure picture.

Step-by-step for each row:

- Read the archive name (1947 original) on the left.
- Look at the structure picture on the right.
- Ask: does this structure match what the archive name describes?
- If yes -> Confirm OK. If wrong and you know the fix -> Override SMILES. If related but not exact -> PARTIAL_MATCH. If clearly wrong -> Reject.
- Click Apply. The row is marked Reviewed and saved - do not review it again.
- Use Next unreviewed to move on.

2. Review queues (priority order)

Your team lead may assign queues. Default priority:

Queue 1 - NEEDS_REVIEW (~537 rows)

SMARTS failed. Main workload. Required.

Queue 2 - CCAST CONVERT_FAILED (~127)

No IUPAC generated. Review structure anyway.

Queue 3 - CCAST SUSPICIOUS (~293)

Low name similarity. Optional spot-check.

Queue 4 - CCAST APPROVED (~1,044)

SMARTS PASS + CCAST OK + Pass 1 + similarity ≥ 0.55 . Reference only.

Queue 5 - PARTIAL_MATCH / Failed

Known incomplete. Usually no action needed.

3. Rules for reviewers

- Always enter your real name before reviewing.
- Do not change rows already marked Reviewed by someone else.
- When unsure, add a note and ask your team lead rather than guessing.
- Trust the structure picture and archive name more than automated flags.
- CCAST OK does not mean SMARTS FAIL rows are correct - still review them.
- Similarity scores are text comparisons only - not proof of structure correctness.

4. Glossary - terms and flags

The web app shows (i) icons next to fields with short explanations. This section has the full descriptions.

Why we are reviewing

The 1947 King USDA dataset lists chemical names tested as insect repellents. Modern machine-learning needs structures, not just names. We used computer programs to look up each name and assign a SMILES code (a line notation for a molecular structure). Automated checks catch many errors but not all. Human reviewers confirm, correct, or reject each assignment before the data is used for modeling.

Archive name (1947 original)

The archive name is stored in the Chemical column. It is the wording from the 1947 King USDA repellent study. Spelling, punctuation, and archaic naming (e.g. old trivial names or abbreviations) are preserved. When you review a row, ask: does the drawn structure reasonably correspond to this historical name?

Resolved name

During Pass 1 and Pass 2, we sent the archive name to public chemistry databases (mainly CIRpy / PubChem-style lookups). The resolved name is what those services returned as their best match. A different spelling does not always mean a wrong structure - but a completely unrelated name is a red flag. Compare resolved name, archive name, and the structure picture together.

SMILES

SMILES (Simplified Molecular Input Line Entry System) is a standard way to write a structure as text, e.g. CCO for ethanol. If the SMILES is wrong, every downstream check (SMARTS, IUPAC, modeling) will be wrong too. When in doubt, use Override SMILES only if you are confident in the correction, or Reject if no reliable structure exists.

Pass 1

Pass 1 sent each archive name to chemical identifier services with no extra manipulation. Status RESOLVED_PASS1 means a structure was found on the first try. These are generally the most trustworthy automatic hits, though some still need SMARTS or manual review.

Pass 2

Pass 2 tried normalized spellings, fragments, and alternate forms of names that failed Pass 1. About 637 structures were found this way. Validation_Risk may show HIGH, MEDIUM, or LOW. HIGH-risk Pass 2 hits were later downgraded to PARTIAL_MATCH in a correction step. Treat Pass 2 rows with more skepticism than Pass 1.

Pass 3

Pass 3 does not usually find new names from scratch. It checks structures we already have. Step 1 (local): SMARTS rules test whether the structure contains chemical groups suggested by the name; OPSIN/ChemSpider may rescue a few failures. Step 2 (CCAST GPU): converts SMILES back to a systematic IUPAC name and compares it to the archive name. Pass3_Status summarizes the outcome (VALIDATED_OK, NEEDS_REVIEW, etc.).

SMARTS check

SMARTS is a language for describing substructures within a molecule. We wrote rules such as 'if the name says alcohol, the structure should contain an -OH group.' PASS: rule ran and matched. FAIL: rule ran and did not match - review required. SKIP: no rule applied (not proof of correctness). PARSE_ERROR: SMILES could not be read. NEEDS_REVIEW queue is mostly SMARTS FAIL rows.

IUPAC name (reverse-generated)

IUPAC names are standardized systematic chemical names. On CCAST we used a machine-learning model (chemical-converter) to go from SMILES to IUPAC - the reverse of naming. The result is a sanity check, not a ground-truth label. Old 1947 trivial names will often look very different from modern IUPAC strings even when the structure is correct.

Similarity score (Jaro-Winkler)

Jaro-Winkler compares the archive name to the generated IUPAC name (and sometimes to the resolved name). Scores near 1.0 mean similar spelling; scores below ~0.45 trigger SUSPICIOUS or LOW_SIMILARITY flags. Shared words can inflate scores even when structures are wrong - always prioritize the structure drawing and archive name over this number.

CCAST flag: OK

OK means the reverse naming step produced an IUPAC string and the Jaro-Winkler score was not in the lowest band. Many OK rows are fine, but hundreds of SMARTS FAIL rows also show CCAST OK because shared words in names can score well. Never use OK alone to skip review when SMARTS failed.

CCAST flag: SUSPICIOUS

SUSPICIOUS indicates the generated IUPAC name does not resemble the 1947 archive name very closely. This can mean a wrong structure, an archaic trivial name, or both. Open the structure image and judge whether it fits the archive name.

CCAST flag: CONVERT_FAILED

CONVERT_FAILED means the reverse naming model returned nothing - often due to CUDA errors or difficult structures. The SMILES may still be valid. Review the structure on its own merits; do not assume failure because IUPAC is missing.

CCAST flag: LOW_SIMILARITY

LOW_SIMILARITY is the strictest text-based flag. Treat these like SUSPICIOUS but with even more caution. Only a handful of rows have this flag in the current dataset.

CCAST APPROVED (unquestionable tier)

CCAST APPROVED is our strict gold-standard tier (~1,044 rows). A row must meet ALL of: (1) Pass3_CCAST_Flag is OK, (2) SMARTS result is PASS (not SKIP or FAIL), (3) Status is RESOLVED_PASS1 (direct first-pass lookup, not Pass 2 retry), and (4) Jaro-Winkler similarity between archive name and IUPAC is at least 0.55. Show these to reviewers as examples of good matches. They are not in the mandatory review queue. CCAST OK alone is weaker - hundreds of SMARTS-fail rows also show CCAST OK.

VALIDATED_OK

VALIDATED_OK means SMARTS returned PASS or SKIP (no FAIL). These ~4,114 rows are the lowest-priority queue. You may browse them to spot obvious mistakes, but focus first on NEEDS_REVIEW. SKIP rows were not proven correct - only that no rule flagged them.

NEEDS_REVIEW

NEEDS_REVIEW marks SMARTS FAIL rows (~537). This is the main workload. For each row, compare archive name, resolved name, and structure. Choose Confirm OK if you accept the SMILES, Override if you have a correction, Mark PARTIAL_MATCH if close but not exact, or Reject to clear the structure.

PARTIAL_MATCH

PARTIAL_MATCH rows (~526) are known ambiguous or high-risk matches from Pass 2 corrections. They are kept for transparency but should not be used as confident structures in models unless manually corrected.

FAILED_BOTH_PASSES

About 1,596 names could not be resolved automatically. Some received a Pass 3 rescue structure (RESOLVED_PASS3). Rows with no SMILES have nothing to draw - browse only.

Reviewed / Not reviewed

Each reviewed row records who reviewed it, when, and what action they chose. Reviewed rows cannot be changed by other reviewers. Enter your name in the sidebar before reviewing. Progress is saved to the project repository after each review.

Action: Confirm OK

Use Confirm OK when the structure picture matches the 1947 name and you see no obvious error. This marks the row as curated and ready for modeling tiers.

Action: Mark PARTIAL_MATCH

Use when the structure is in the right chemical family but wrong form, salt, stereochemistry, or isomer. Downgrades confidence for modeling.

Action: Reject

Reject when the structure is clearly wrong and you cannot supply a better SMILES. The row is excluded from the curated modeling set.

Action: Override SMILES

Use only when you are confident in the correction (e.g. from a trusted database or expert knowledge). Paste the new SMILES in the override box, then apply.

5. Reference - column names in the app

Field	Meaning
Chemical	Archive name from 1947 dataset
Resolved_Name	Name returned by database lookup (Pass 1/2)
SMILES	Structure code; drives the picture
Status	Pass 1/2 outcome (RESOLVED_PASS1, PARTIAL_MATCH, etc.)
Pass3_Status	VALIDATED_OK, NEEDS_REVIEW, CURATED_OK after you review
Pass3_SMARTS_Result	PASS, FAIL, or SKIP
Pass3_SMARTS_Reason	Why SMARTS passed or failed
STOUT_IUPAC	IUPAC name generated from SMILES on CCAST
JaroWinkler_Score	Similarity 0-1 vs archive name
Pass3_CCAST_Flag	OK, SUSPICIOUS, CONVERT_FAILED, LOW_SIMILARITY
Validation_Risk	Pass 2 risk level (HIGH / MEDIUM / LOW)

6. Questions?

Contact your team lead (Insectapel / NDSU) if a name is ambiguous, the app will not save, or you think a reviewed row needs to be reopened.